

Aim:

You are given a dataset in the form of a dictionary containing examination records of 10 students. The dataset includes their names, scores, number of attempts, and whether they qualified or not. You need to create a pandas dataset frame for the given data:

Dataset Details:

- **name:** Name of the student.
- **score:** Exam score (float, may contain missing values, 'NaN').
- **attempts:** Number of exam attempts (integer).
- **qualify:** Whether the student is qualified ('yes' or 'no').

The data frame is indexed using custom labels from 'a' to 'j'.

Objectives:

You need to convert the given data into a Pandas data frame and then perform the following operations using appropriate Pandas methods:

- **Print Original Dataset:** Display the original DataFrame.
- **Select Name and Score Columns:** Select and print the `name` and `score` columns.
- **Select Students with Score Between 15 and 20:** Select and print rows where the score is between 15 and 20 (inclusive).
- **Identify Missing Scores:** Print all rows where the score is missing (NaN).
- **Update Score for Student with Label 'd':** Update the Score for the Student with Index Label 'd' to 11.5 and display the updated DataFrame.
- **Calculate Total Exam Attempts:** Calculate and print the total number of exam attempts by all students.

Task: Ensure that missing values are replaced with 0 after identifying missing scores but before updating and calculating totals.

Instructions:

1. Create a Pandas DataFrame using the provided `exam_data` and labels.
2. Perform the tasks in the order listed above.
3. Use appropriate Pandas methods for:
 - Selecting columns and rows
 - Filtering by conditions
 - Identifying missing values
 - Updating values
 - Calculating totals

Note:

- A data frame is already provided to you in the editor; perform the required operation to achieve the task.
- Refer to the visible test cases for better understanding and to match the printing pattern.

Source Code:

```
exam_analysis.py
```

```
import pandas as pd
import numpy as np

exam_data = {
```

```

ti', 'Vishal', 'Pranay'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']
}
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

    # write your code here...
df=pd.DataFrame(exam_data, index=labels)
def select_name_score(df):
    return df[['name','score']]
def select_score_range(df):
    return df[df['score'].between(15,20)]
def select_missing_scores(df):
    return df[df['score'].isna()]
def update_score_d(df):
    df.loc['d','score']=11.5
    return df
def total_attempts(df):
    return df['attempts'].sum()
print("Original DataFrame:")
print(df)

print("Selected 'name' and 'score' columns:")
print(select_name_score(df))

print("Rows with scores between 15 and 20:")
print(select_score_range(df))

print("Rows with missing scores:")
print(select_missing_scores(df))

print("DataFrame after updating score in row 'd' to 11.5:")
df_updated = update_score_d(df.copy())
print(df_updated)

print("Total attempts by all students:")
print(total_attempts(df))

```

Execution Results - All test cases have succeeded!

Test Case - 1				
User Output				
Original DataFrame:				
	name	score	attempts	qualify
a	Ajay	12.5	1	yes
b	Vijay	9.0	3	no
c	Vivek	16.5	2	yes
d	Atharv	NaN	3	no
e	Apurva	9.0	2	no
f	Rupal	20.0	3	yes
g	Priyal	14.5	1	yes
h	Shrushti	NaN	1	no

i	Vishal	8.0	2	no
j	Pranay	19.0	1	yes
Selected 'name' and 'score' columns:				
	name	score		
a	Ajay	12.5		
b	Vijay	9.0		
c	Vivek	16.5		
d	Atharv	NaN		
e	Apurva	9.0		
f	Rupal	20.0		
g	Priyal	14.5		
h	Shrushti	NaN		
i	Vishal	8.0		
j	Pranay	19.0		
Rows with scores between 15 and 20:				
	name	score	attempts	qualify
c	Vivek	16.5	2	yes
f	Rupal	20.0	3	yes
j	Pranay	19.0	1	yes
Rows with missing scores:				
	name	score	attempts	qualify
d	Atharv	NaN	3	no
h	Shrushti	NaN	1	no
DataFrame after updating score in row 'd' to 11.5:				
	name	score	attempts	qualify
a	Ajay	12.5	1	yes
b	Vijay	9.0	3	no
c	Vivek	16.5	2	yes
d	Atharv	11.5	3	no
e	Apurva	9.0	2	no
f	Rupal	20.0	3	yes
g	Priyal	14.5	1	yes
h	Shrushti	NaN	1	no
i	Vishal	8.0	2	no
j	Pranay	19.0	1	yes
Total attempts by all students:				
19				